2024

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : SEC-3P

(Mobile App Development)

Full Marks : 50

The figures in the margin indicate full marks.

SET - V

Marks Distribution :

User Interface (UI) / Front End	:	05
Procedure/Steps	:	10
Source Code	:	10
Output	:	07
Conclusion/Discussion	:	03
Experiment Total	**:**	**35**
Laboratory Notebook	:	05
Viva voce	:	10
Total	**:**	**50**

Answer ***any one*** question.

1. Develop a simple Flutter app that calculates simple interest. There will be textboxes for entry of Principal amount, Rate of Interest and a dropdown box for input of Tenures in years.
   On clicking button "Get Interest" the simple interest value will be printed on a label. Keep validations for non null able numeric entry. There will be a caption 'Simple Interest Calculation' on app bar. 35

2. Develop a simple Flutter app that converts values in inch, foot to centimeter. There will be two dropdowns for entries of inch and foot value, a convert button on clicking the result will be shown in two different labels as '1 inch = 2.54 cm' and '1 ft = 30.48 cm'. Keep validations for non null able entry. Display 'Converter application' title on App bar. 35

3. Develop a Flutter app that will have a form on 'Issue of Book' in a library. User inputs Book name, Author, Membership No, Date of Issue. On pressing 'Submit' button 'Date of Return' will be displayed in another text field adding 10 days to date of issue. Also showing message 'Book Issued to <Membership No>'. The app will also check valid non null entries. The app bar will display the title 'Book Issue Form' of the app. 35

4. Develop a flutter app that inputs a enrollment number in a text field and print the roll number in word. Keep checking for non null and numeric entry. The app bar will display the title 'Number in Word' of the app.

   Eg : input number : 23401

   Output In word : Two Three Four Zero One. 35

5. Develop a Flutter app that inputs a sentence and change the sentence to upper case. Also check if the length of the string is more than 12 or not. The sentence may contain alphabets, numbers, comma etc. The app will provide a test box to take input sentence and labels to display the results. Give proper title on the app bar. 35

6. Develop a Flutter app that displays a digital clock as HH:MM:SS AM/PM (12 hrs) and today's date in 'DD-MM-YYYY' format. Also keep two options named Pink and Blue, the background will be changed as users selection. The app bar must show a title of the page. 35

7. Develop a simple Flutter app that generates result of a student. User inputs name of student, semester, marks in paperl, paper2 and paper3, calculate Paper grade points and SGPA on pressing 'Generate Result' button and display in text fields. Full marks is 100. Grade point and SGPA will be calculated as follows taking credit value 2 for each paper. There will be suitable title 'Result' on app bar.

   If the student scores P% where P>=30 his/her grade point will be [3.0 + 0.1 × (P–30)] else print 'semester not cleared'.

   Eg. For P = 40, grade point = 3.0 + 0.1 × (40 – 30) = 4.0.

   SGPA can be calculated by $\Sigma(P_i \times C_i) / \Sigma C_i$, where i = 1...n, n = No. of papers, P_i denotes grade point, C_i denotes credit of each paper. 35

8. Develop a Flutter app that includes two images, utilizes necessary widgets so that upon pressing a button it creates a smooth transition between the two images. 35